A Machine-Checked Audit of the Extended-Extraction RG Contraction in the Wilson-Lattice / OS Route to a 4D Yang–Mills Mass Gap

The MeTTapedia formalization effort

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Abstract

Ben Goertzel's manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14) proposes to construct a continuum $\mathrm{SU}(N)$ Yang–Mills theory as a limit of Wilson lattice gauge theory, organized around a multiscale renormalization-group (RG) bootstrap, a Koteckš–Preiss (KP) polymer expansion yielding exponential clustering at a fixed physical scale, and an Osterwalder–Schrader (OS) reconstruction producing a Hamiltonian with a strictly positive spectral gap. The technical keystone is an *extended-extraction* RG contraction $\Lambda = C_1 \lambda_{\mathrm{irr}} < 1$, with an advertised recombination constant $C_1 \le 2224$ and an irrelevant-eigenvalue factor $\lambda_{\mathrm{irr}} = 2^{-13}$ (block factor $b = 2$, extraction depth $d_{\max} = 16$), so that $2224 \cdot 2^{-13} \approx 0.27 < 1$. We report what the MeTTapedia formalization effort has machine-checked in Lean about this route. We model the contraction predicate `HasExtendedExtractionContraction`, verify the knife-edge arithmetic ($2224 \cdot 2^{-13} < 1$ holds at $d_{\max} = 16$; the analogous bound *fails* at the nearest even depth $d_{\max} = 14$, where $2^{11} = 2048 < 2224$), establish genuine finite $\mathbb{Z}_2$ and $\mathbb{Z}_3$ single-link strong-coupling spectral gaps through a finite OS reconstruction, and *gate the continuum endpoint as open*: the continuum mass gap carries an explicit open status with no claimed support. We are precise about scope. The checked arithmetic takes the recombination constant $C_1 = 2224$ as a *given input*. The load-bearing quantity — whether the true extraction-projection operator norm, hence C_1, is as small as advertised — lies outside the present formalization; we record neutrally that the manuscript's own written expression for that norm appears to grow rapidly with the extraction depth $d_{\max}$, so the small constant is not self-evidently justified, and we flag this as the key unformalized risk rather than asserting any resolution. The finite gaps that are checked are real but Clay-irrelevant (abelian, single-link, strong-coupling, finite). The Yang–Mills mass gap itself remains open; we make no claim about it.

1 The problem and Ben Goertzel's route

1.1 The Millennium problem and the lattice strategy

Fix the gauge group $G = \mathrm{SU}(N)$ with $N \ge 2$ and let $\mathbb{R}^4$ be Euclidean space. The Yang–Mills mass-gap problem asks for the construction of a nontrivial continuum quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group G, satisfying the Wightman / Osterwalder–Schrader axioms, whose reconstructed Hamiltonian has a unique vacuum and a strictly positive spectral gap on the orthocomplement of the vacuum.

Goertzel's v14 manuscript follows Wilson's lattice regularization. One discretizes $\mathbb{R}^4$ as a hypercubic lattice $a\mathbb{Z}^4$ with spacing $a > 0$, with link variables $U_{x,\mu} \in G$ on oriented nearest-neighbor links. The plaquette variable is the parallel transporter around an elementary square,

$$U_{x,\mu\nu} \;=\; U_{x,\mu}\, U_{x+ae_\mu,\nu}\, U^{-1}_{x+ae_\nu,\mu}\, U^{-1}_{x,\nu}, \qquad 1 \le \mu < \nu \le 4,$$

and the Wilson action penalizes deviation of plaquettes from the identity,

$$S_{\Lambda,a}(U) \;=\; \beta \sum_{p\subset\Lambda} V(U_p), \qquad \beta = \frac{2N}{g^2}, \qquad V(U) := 1 - \tfrac{1}{N}\mathrm{Re}\,\mathrm{Tr}(U) \in [0,2].$$

Weak coupling is $\beta \gg 1$. The strategy is: (Step 1) start from the well-defined finite-dimensional lattice integral; (Step 2) run a multiscale RG bootstrap controlling the effective theory across all scales; (Step 3) establish exponential clustering at a fixed physical scale ℓ_* via a KP polymer expansion; (Step 4) transfer clustering from the coarse scale back to the fine scale with a glue / back-propagation lemma; (Step 5) reconstruct the continuum Hilbert space, vacuum, and Hamiltonian by OS, with the clustering rate becoming the spectral gap; (Step 6) verify nontriviality through a nonzero third cumulant of the action density.

1.2 The continuum-limit obstruction and extended extraction

In any RG scheme one must show that the "irrelevant" sector (operators of canonical dimension > 4) contracts at each step:

$$\|K^{\mathrm{irr}}_{j+1}\| \;\le\; \Lambda\, \|K^{\mathrm{irr}}_j\| + (\text{source terms}), \qquad \Lambda < 1.$$

The manuscript identifies a sharp obstruction in the *naive* scheme, where one extracts only the relevant and marginal operators of dimension ≤ 4. Under rescaling, a dimension-d operator scales by b^{4-d}, so the leading irrelevant operators (dimension 6) contract by $b^{-2} = 0.25$ for $b = 2$. But the RG step also recombines polymers, contributing a combinatorial constant C_1. The manuscript's rigorous worst-case bound is $C_1 \le 2224$ for $b = 2$, giving

$$\Lambda_{\mathrm{naive}} \;=\; C_1\, b^{-2} \;\ge\; 2224 \times 0.25 \;=\; 556 \;>\; 1,$$

an *expanding* map: initial smallness would grow like 556^J over $J \sim |\log a_0|$ steps and diverge as $a_0 \to 0$.

The proposed resolution is *extended extraction.* Instead of extracting only dimension ≤ 4 operators, one extracts all gauge-invariant local operators up to canonical dimension $d_{\max} = 16$ (for $b = 2$). The irrelevant sector then begins at dimension $d_{\max} + 1 = 17$ and contracts by

$$\lambda_{\mathrm{irr}} \;=\; b^{4-(d_{\max}+1)} \;=\; b^{3-d_{\max}} \;=\; 2^{-13} \;\approx\; 1.2 \times 10^{-4}.$$

With the same rigorous $C_1 \le 2224$, the effective contraction factor becomes

$$\boxed{\Lambda \;=\; C_1\, \lambda_{\mathrm{irr}} \;\le\; 2224 \cdot 2^{-13} \;\approx\; 0.27 \;<\; 1.}$$

The manuscript presents this as genuine contraction "with substantial slack." The price is tracking a finite-dimensional vector of higher couplings $\vec{g}_j = (g_j^{(4)}, g_j^{(6)}, \ldots, g_j^{(d_{\max})})$ rather than the single gauge coupling; the manuscript argues these are perturbatively small and do not affect the universal one-loop beta-function coefficient $b_0 = 11/(3{\cdot}16\pi^2)$, which is determined entirely by local ultraviolet information.

1.3 The recombination constant and the extraction-projection norm

The constant C_1 is itself a product of factors (manuscript Computation O.7),

$$C_1 \;\le\; C_{\text{fluct}} \cdot C_{\text{cumulant}} \cdot C_{\text{recomb}} \cdot C_{\text{extract}} \;\le\; 1.65 \times 3b^4 \times 14 \times 2 \;\approx\; 139\, b^4,$$

which for $b = 2$ gives $C_1 \le 139 \times 16 \approx 2224$. The last factor $C_{\text{extract}} = \|P_{\le d_{\max}}\|_{\text{op}}$ is the operator norm of the extraction projection onto the finite-dimensional space of operators of dimension $\le d_{\max}$. The manuscript bounds it (its Computation O.7(d)) by

$$\|P_{\le d_{\max}}\|_{\text{op}} \;\le\; \sum_{d=0}^{d_{\max}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \;\le\; 1 + O\big((r_0/r_1)^{d_{\max}}\big),$$

and asserts $\|P_{\le d_{\max}}\|_{\text{op}} \le 2$ for $r_1/r_0 = 1/2$ and $d_{\max} = 16$. We return to this expression in §3; it is the load-bearing input on which the smallness of C_1, and hence the contraction, rests.

2 What we formalized

The formalization lives in the Lean namespace `Mettapedia.QuantumTheory.YangMills`. We summarize the four checked components: the contraction predicate, the knife-edge arithmetic, the finite strong-coupling gaps with their finite OS reconstruction, and the continuum-open gating. All theorem and definition names below are reproduced verbatim from the corresponding modules.

2.1 The extended-extraction contraction predicate

The module `Mettapedia.QuantumTheory.YangMills.RGBootstrap` models the route's contraction obligation as a predicate over a recombination constant C, block factor b, and extraction depth $d_{\max}$. The irrelevant scaling factor is $b^{3-d_{\max}}$ (an integer power, since the useful cases have $d_{\max} > 3$), and a contraction is a nonnegative constant whose product with the irrelevant scale is below 1.

```
/-- Irrelevant scaling factor ‘b^(3-dmax)‘ used by the extended-extraction
discussion.  The exponent is an integer because the useful cases have
‘dmax > 3‘. -/
def irrelevantScale (b dmax : ℕ) : ℝ :=
  (b : ℝ) ^ (3 - (dmax : ℤ))

/-- The route-level contraction audit for a combinatorial constant and an
irrelevant scaling factor. -/
def HasRGContraction (C scale : ℝ) : Prop :=
  0 ≤ C ∧ 0 ≤ scale ∧ C * scale < 1

/-- The extended-extraction audit specialized to ‘λ = b^(3-dmax)‘. -/
def HasExtendedExtractionContraction (C : ℝ) (b dmax : ℕ) : Prop :=
  HasRGContraction C (irrelevantScale b dmax)
```

A threshold lemma in the same module records the exact content of the predicate: for $b > 0$ and $d_{\max} \ge 3$, the contraction `HasExtendedExtractionContraction C b dmax` holds if and only if $0 \le C < b^{d_{\max}-3}$. Thus the entire arithmetic content of the route is the comparison of C with the threshold $b^{d_{\max}-3}$.

2.2 The knife-edge arithmetic

The module machine-checks that the advertised constant contracts at $d_{\max} = 16$ and that the exact one-step gain is $2224 \cdot 2^{-13} = 139/512 < 1/3$:

```
/-- The advertised numerical slack: ‘2224 * 2^-13 < 1‘. -/
theorem rgContraction_2224_two_sixteen :
    HasExtendedExtractionContraction 2224 2 16 := by
  refine ⟨by norm_num, by rw [irrelevantScale_two_sixteen]; norm_num, ?_⟩
  rw [irrelevantScale_two_sixteen]
  norm_num

/-- Exact advertised one-step gain for the extended-extraction constants. -/
theorem rgGain_2224_two_sixteen_eq :
    (2224 : ℝ) * irrelevantScale 2 16 = (139 : ℝ) / 512 := by
  rw [irrelevantScale_two_sixteen]
  norm_num
```

Crucially, the same constant does *not* contract at the nearest even extraction depth below the advertised one. At $d_{\max} = 14$ the threshold is $2^{11} = 2048 < 2224$, so the contraction is arithmetically false; the module proves that any contraction at $d_{\max} = 14$ would force $C < 2048$, and hence that 2224 fails:

```
/-- Any contraction proof at ‘dmax = 14‘ must improve the linear recombination
constant below ‘2048‘.  The manuscript’s advertised bound ‘2224‘ is therefore
insufficient for this lower extraction depth. -/
theorem rgContraction_two_fourteen_forces_constant_lt_2048
    {C : ℝ} (h : HasExtendedExtractionContraction C 2 14) :
    C < 2048 := by
  have hprod : C * irrelevantScale 2 14 < 1 := h.2.2
  rw [irrelevantScale_two_fourteen] at hprod
  nlinarith

/-- Same-constant blocker at the nearest even extraction depth below ‘16‘. -/
theorem not_rgContraction_2224_two_fourteen :
    ¬ HasExtendedExtractionContraction 2224 2 14 := by
  intro h
  have hC : (2224 : ℝ) < 2048 :=
    rgContraction_two_fourteen_forces_constant_lt_2048 h
  norm_num at hC
```

This is the knife edge: the choice $d_{\max} = 16$ is exactly what makes the advertised constant contract, and the manuscript's own table shows $d_{\max} \in \{10, 12, 14\}$ all fail against $C_1 \leq 2224$ while $d_{\max} = 16$ succeeds. The module `Mettapedia.QuantumTheory.YangMills.RGCrux` packages this into a route-level statement: a mass-gap route that keeps the advertised constant 2224 but uses an even extraction depth strictly below 16 cannot exist, since the required contraction is arithmetically false before any OS / clustering layer is reached.

```
/-- Same-constant lower-extraction certificate in the manuscript’s even
canonical-dimension shape: the cutoff is strictly below ‘16‘, is even, and uses
the same advertised recombination bound ‘2224‘. -/
def SameConstantEvenBelowSixteenRGCertificate (dmax : ℕ) : Prop :=
  dmax < 16 ∧ Even dmax ∧ HasExtendedExtractionContraction 2224 2 dmax
```

```
/-- No same-constant RG certificate exists for an even manuscript-shape cutoff
strictly below `16`. -/
theorem not_sameConstantEvenBelowSixteenRGCertificate
    {dmax : ℕ} :
    ¬ SameConstantEvenBelowSixteenRGCertificate dmax := by
  intro hcert
  have hdmax : dmax ≤ 14 :=
    even_lt_sixteen_le_fourteen hcert.1 hcert.2.1
  exact (not_rgContraction_2224_two_of_dmax_le_fourteen hdmax) hcert.2.2
```

A strengthened companion theorem, `not_atLeast2048EvenBelowSixteenRGCertificate`, shows that any even lower-extraction repair below $d_{\max} = 16$ would need a genuinely sharper constant ($C < 2048$), not merely the advertised one. We stress that this is an honest delimiter of the arithmetic surface, not a refutation of the route: at the raw natural number $d_{\max} = 15$ the same constant *does* contract ($2^{12} = 4096 > 2224$), and the module records this explicitly so that the obstruction is read precisely as an even canonical-dimension statement matching the manuscript's dimensions $4, 6, \ldots, d_{\max}$.

2.3 Finite strong-coupling spectral gaps and finite OS reconstruction

The module `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction` formalizes the finite algebraic core of OS reconstruction: a reflection-positive finite kernel induces a positive pairing on observables, and a transfer operator that is self-adjoint for that pairing descends to the pairing quotient and remains self-adjoint there. On this base, the modules `Z2StrongCouplingGap` and `Z3StrongCouplingGap` build explicit finite single-link abelian lattice gauge models. For $\mathbb{Z}_2$, the one-link heat-kernel transfer at heat time 1 has eigenvalues 1 and $\exp(-1)$, so the dimensionless gap is $-\log(\exp(-1)/1) = 1$:

```
theorem z2StrongCouplingGap_eq_one :
    z2StrongCouplingGap = 1 := by
  simp [z2StrongCouplingGap, z2StrongCouplingLambdaTwo,
    z2StrongCouplingLambdaOne, z2StrongCouplingRatio, Real.log_exp]

/-- End-to-end finite lattice theorem: the explicit `Z₂` strong-coupling model is
nontrivial, reflection positive, has an OS transfer operator, and has
dimensionless transfer gap at least `1`. -/
theorem z2StrongCoupling_latticeYangMills_massGap_landmark :
    ∃ _ : Z2StrongCouplingGapCanary,
      (∃ x y : Z2OneLinkConfig, x ≠ y) ∧
        OSReflectionPositive z2StrongCouplingKernel ∧
        z2StrongCouplingGap = 1 ∧
        (1 : ℝ) ≤ z2StrongCouplingGap ∧
        0 < z2StrongCouplingGap := by
  exact
    ⟨z2StrongCouplingPositiveCanary,
      z2OneLinkConfig_nontrivial,
      z2StrongCouplingKernel_reflectionPositive,
      z2StrongCouplingGap_eq_one,
      z2StrongCouplingGap_lower_bound_one,
      z2StrongCouplingGap_positive⟩
```

The $\mathbb{Z}_3$ module proves the analogous landmark with two mean-zero transfer eigenvectors and the same unit gap. These theorems are unconditional and non-vacuous: they exhibit genuine reflection positivity, a genuine finite OS transfer reconstruction, and a strictly positive spectral gap. They are also kept rigorously separate from any continuum claim. Each module carries an explicit scaling diagnostic showing that a finite gap need not survive a naive continuum schedule. For the heat-time schedule $t(a) = a^2$, the *physical* gap (dimensionless gap divided by lattice spacing) closes as $a \to 0$:

```
/-- A concrete gap-closing schedule for the finite model under the heat-time
scaling ‘t(a)=a^2‘: the physical gap can be made smaller than any positive
threshold by taking the lattice spacing small enough. -/
theorem z2QuadraticHeatTime_physicalGap_closes :
    ∀ ε : ℝ, 0 < ε →
      ∃ a : ℝ,
        0 < a ∧ a < ε ∧
          z2HeatTimePhysicalGap a (a ^ 2) < ε := by
  intro ε hε
  refine ⟨ε / 2, by linarith, by linarith, ?_⟩
  have hne : ε / 2 ≠ 0 := by linarith
  rw [z2QuadraticHeatTime_physicalGap_eq hne]
  linarith
```

2.4 The continuum endpoint is gated open

The module `Mettapedia.QuantumTheory.YangMills.ProofState` records a status ledger over the route's nodes. The finite RG / extraction surfaces are marked `checked` or `refuted` as appropriate, but the continuum mass-gap endpoint carries an explicit `openGoal` status, gated behind a constructive-QFT promotion barrier (continuum measure, Euclidean covariance, reflection positivity, OS reconstruction, and Hamiltonian mass-gap transfer — none of which is represented for the RG / extraction route, which supplies only a finite ledger):

```
/-- The final mass-gap endpoint remains an open goal in this lane. -/
def yangMillsMassGapEndpointNode : YangMillsProofNode where
  id := "yang-mills.mass-gap-endpoint"
  status := .openGoal
  truthValue := ⟨0, 95⟩
  evidence := "currentYangMillsMassGapEndpoint_blockedByConstructiveGate blocks promotion
     from the audited RG/extraction surfaces to a continuum mass-gap endpoint ..."
  nextObligation := "Connect the RG and extraction surfaces to a genuine continuum mass-
    gap statement only after the constructive-QFT gate, OS/Hamiltonian transfer
    dependency packet, challenge-world interface packet, and mass-gap promotion gate
    clear."

theorem yangMillsMassGapEndpointNode_open :
    yangMillsMassGapEndpointNode.status = .openGoal := by
  rfl
```

A companion theorem `currentYangMillsMassGapEndpoint_blockedByConstructiveGate` records that the endpoint is open, the constructive checks are all unrepresented, and the only supplied interface is the finite RG / extraction ledger. In short, the formalization does not overclaim: nothing in it asserts a continuum mass gap.

3 Honest status and scope

This lane is, by construction, conservative; we now state precisely what is and is not established.

The contraction constant is an input, not a conclusion. Every arithmetic theorem above takes the recombination constant as a *given real number*. The predicate `HasExtendedExtractionContraction C b dmax` is a statement about C, b, $d_{\max}$ alone; the checked facts `rgContraction_2224_two_sixteen` and `not_rgContraction_2224_two_fourteen` verify the comparison $C \lesseqgtr b^{d_{\max}-3}$ for the specific value $C = 2224$. Nothing in the formalization derives that 2224 (or any $O(1) \cdot b^4$ bound) is the correct value of the manuscript's recombination constant. The Lean ledger is therefore faithful to the route's bookkeeping but agnostic about the route's hardest analytic claim.

The extraction-projection norm justification is the key unformalized risk. The smallness of C_1 rests on the factor $C_{\text{extract}} = \|P_{\leq d_{\max}}\|_{\text{op}}$, which the manuscript bounds by

$$\|P_{\leq d_{\max}}\|_{\text{op}} \;\leq\; \sum_{d=0}^{d_{\max}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \;\leq\; 1 + O\big((r_0/r_1)^{d_{\max}}\big),$$

asserting the value ≤ 2 for $r_1/r_0 = 1/2$, $d_{\max} = 16$. We record neutrally a tension internal to this written expression: with $r_1/r_0 = 1/2$ the summand $(r_1/r_0)^{-d} = 2^d$ *grows* geometrically in d, and the binomial weights only enlarge it further, so the written partial sum is an *increasing* function of $d_{\max}$ that grows at least like $2^{d_{\max}}$; the asserted closed form $1 + O\big((r_0/r_1)^{d_{\max}}\big) = 1 + O(2^{-d_{\max}})$ instead behaves as if the summand were $(r_1/r_0)^d = 2^{-d}$, i.e. as if it *decayed*. We do not adjudicate which reading is intended. We observe only that this single step decides whether C_1 stays $O(1) \cdot b^4 \approx 2224$ or instead grows with the extraction depth $d_{\max}$ — and that, because the irrelevant factor is $\lambda_{\text{irr}} = 2^{3-d_{\max}}$ while C_{extract} enters C_1 multiplicatively, a C_{extract} that grew like $2^{d_{\max}}$ would overwhelm the very 2^{-13} contraction the scheme relies on. This is precisely the part not captured in Lean. We flag it as the key unformalized risk; we do not assert that the constant is wrong, only that its smallness is not self-evident from the manuscript's own stated expression and is not yet machine-checked.

The finite gaps are real but Clay-irrelevant. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectral gaps are honest theorems, but they are abelian, single-link, strong-coupling, and finite. They exemplify the finite OS-reconstruction mechanism; they are not lattice approximations to a continuum $\mathrm{SU}(N)$ theory, and the accompanying scaling diagnostics show that the physical gap can close under a naive continuum schedule. They establish the machinery, not the target.

The continuum mass gap is correctly open. The continuum endpoint is gated `openGoal` with no claimed support, and the constructive-QFT prerequisites are all marked unrepresented. The formalization makes no claim, in either direction, about the existence of a continuum Yang–Mills mass gap.

4 What remains

To turn this lane into evidence about the route's central claim rather than its bookkeeping, the decisive next step is to formalize, or independently bound, the extraction-projection operator norm

$\|P_{\le d_{\max}}\|_{\mathrm{op}}$ as a function of $d_{\max}$, and thereby determine whether the recombination constant C_1 is genuinely $O(1) \cdot b^4$ or grows with the extraction depth. Until that input is established, the checked contraction arithmetic — correct as it is — rests on an assumed value. Beyond it lie the genuinely infinite-dimensional layers the route requires and the present ledger leaves unrepresented: the KP polymer expansion and tree-decay clustering at the stopping scale, the glue / back-propagation transfer of clustering from the coarse to the fine scale, the thermodynamic and continuum OS limits, rotation restoration, the transfer-matrix spectral gap, and the nontriviality witness. Each is an analytic theorem, not an arithmetic comparison, and none is currently machine-checked. The Yang–Mills mass gap remains open.

Lean modules. The results reported here are in `Mettapedia.QuantumTheory.YangMills.RGBootstrap` (the contraction predicate and knife-edge arithmetic), `Mettapedia.QuantumTheory.YangMills.RGCrux` (the even lower-extraction route obstruction), `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction`, `Mettapedia.QuantumTheory.YangMills.Z2StrongCouplingGap`, and `Mettapedia.QuantumTheory.YangMills.Z3StrongCouplingGap` (the finite OS reconstruction and strong-coupling gaps), and `Mettapedia.QuantumTheory.YangMills.ProofState` (the continuum-open gating). The primary mathematical source is Ben Goertzel's manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14).